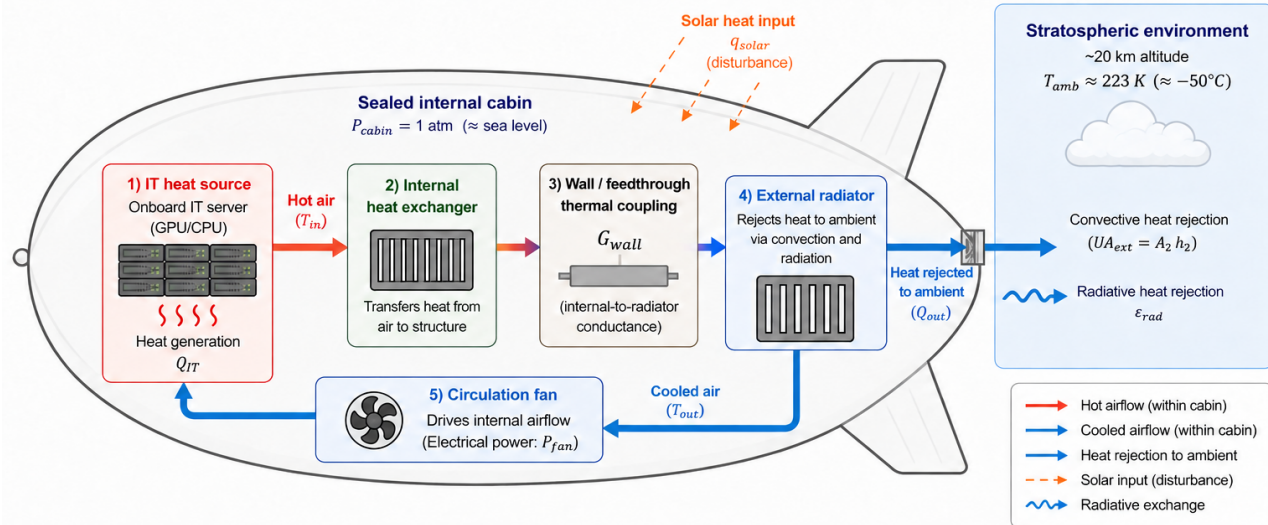


a

Thermal architecture of the HAPS payload

Fan-driven internal air loop with external heat rejection to the stratospheric environment



Thermal paths modeled in this work

Internal air loop and heat exchanger

Heat generated by the IT system is transported by fan-driven internal airflow to a heat-exchanger surface.

$$Q_{IT} \rightarrow \text{air} \rightarrow \text{heat exchanger}$$

Wall / feedthrough coupling

Heat is conducted through an equivalent wall/feedthrough path to the external radiator.

$$Q_{nx} = G_{wall}(T_{hx} - T_{rad})$$

External heat rejection to ambient

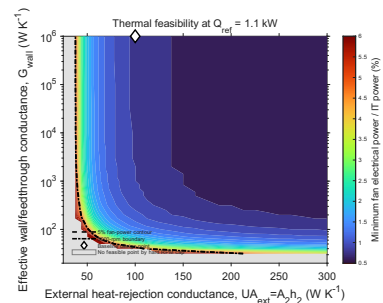
The external radiator rejects heat to the stratospheric environment via convection and radiation.

$$Q_{out} = UA_{ext}(T_{rad} - T_{amb}) + \epsilon_{rad}$$

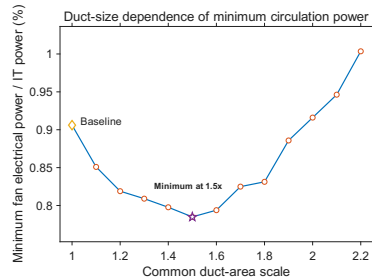
Key outputs for capacity analysis

- Fan power: P_{fan} [W]
- Payload power: $P_{payload} = P_{IT} + P_{fan} + P_{comm}$ [W]
- Cabin temperature: T_{cabin} [$^\circ\text{C}$]
- Thermal feasibility (T_{server} , T_{cabin})

b



c



d

